

ZEENATH ARA SYED

MSc in Robotics and Embedded AI
Ireland • s.zeenath.ara@gmail.com • (+353) 872352726 • [LinkedIn](#)

SUMMARY

Robotics & AI Engineer with 7+ years of software engineering experience, completing MSc dissertation at Maynooth University through an industrial project with St. James Hospital. Skilled in embedded systems, mechanical design (SolidWorks), Unity 3D, ROS2, and computer vision. Developing a medical robotics training platform integrating IMU sensing, tendon actuation, and digital twin visualization. Seeking opportunities in autonomous systems and intelligent control.

INDUSTRIAL PROJECTS

Robotic Eye Movement Simulator for the Dix-Hallpike Test - St. James Hospital(Currently working)

- Currently working on designing system architecture integrating tendon-driven eye mechanism, IMU head tracking (MPU-6050 + Arduino), and Unity 3D digital twin with real-time synchronization (<100ms latency, $\pm 2^\circ$ accuracy).
- Developed mechanical components in SolidWorks, including head housing, tendon routing, and servo-driven eye actuation for multi-axis nystagmus simulation (3-15°/s velocity range).
- Implementing patient profile modeling, automated feedback systems, and TTS/STT interfaces for clinical training scenarios.

ACADEMIC PROJECTS

Pick & Stack 3 colored cubes using Robot Arm and Camera

- Built a robotic arm to pick and stack cubes using ROS2, URDF, and camera perception along with simulation in RViz and Gazebo.
- Developed point-cloud based object detection node and controlled joints via ROS2 publishers with trajectory waypoints; added safety stops and joint-limit handling.
- Designed precise pick-lift-place sequences using camera input and handled with single launch file.

IoT Based WiFi Robot:

- Developed IoT robot to detect hazards using ESP32, Arduino, and ROS2.
- Integrated sensors for real-time monitoring; analyzed data with Python, NumPy, Pandas.
- Enabled wireless control over Wi-Fi in unsafe areas.
- Enhanced safety with real-time alerts, automated responses, and documented ROS2 workflow.

PROFESSIONAL EXPERIENCE

Software Development Engineer - L1, Kongsberg Digital

Apr 2022 - Aug 2025

- Developed a Reasoning Engine called 'Asset Copilot' using LLM models such as GPT4, GPT4o, Llama models and integrated it with the company's product, a Digital Twin.
- Designed and developed a dashboard that evaluates asset copilot performance using various metrics such as precision, fluency, F1 score, hallucination by using prompt engineering techniques.
- Spearheaded the development of the Energy Nomination Dashboard, increasing user engagement by 25%.

Onsite at Shell's Energy Transition Campus, Amsterdam(Nov 2022 - Nov 2023)

- Worked closely with the Shell innovation team in Amsterdam to ideate, prototype, and develop cutting-edge features.
- Developed POC for the product KogniTwin, a Digital Twin in the construction sector.
- Developed POC of an Energy Nomination Dashboard, that gives overview of various energy parameters in real-time.

Software Development Engineer - L2, Kongsberg Digital

July 2021 - Mar 2022

- Collaborated with Customer Success Manager and clients to set team goals.
- Managed and led multiple sub-teams, orchestrated software deployments.
- Implemented core functionalities including 3D implementation and real-time data from InfluxDB and visualize it in the product, KogniTwin.

Software Development Engineer - L3, Kongsberg Digital

July 2020 - June 2021

- Managed and optimized customer data using SQL and MongoDB, improving data integrity and insights.
- Enhanced UI/UX for product usability and visualization using Python and JavaScript.

Software Engineer Trainee, Kongsberg Digital

July 2018 - June 2020

- Led ConEdison project visualizing real-time gas pipeline data on a 3D map; managed millions of data records.
- Delivered end-to-end project delivery, including client interactions and Site Acceptance Testing (SAT).

EDUCATION

MSc - Robotics and Embedded AI

Maynooth University, 2025 - 2026

B.E - Electronics & Communication Engineering

Reva Institute of Technology, 2013 - 2017

CERTIFICATIONS

- Fundamentals of Deep Learning - Nvidia(2026)
- Robotics with Python - Udemy (2025)
- Applied Computer Vision: Object Detection and Recognition - Udemy(2025)
- Professional Scrum Master I - Scrum.org (2024)
- Scrum & SAFe PO/PM - Udemy (2024)

ACHIEVEMENTS

- **MUSE Award - Platinum Level (2026):** Recognized by Maynooth University, the highest recognition for leadership and community engagement.
- **Winner - GenAI Hackathon (2024):** Developed a gas lift optimizer that increased profit margins by 20%.
- **Spotlight Awards (2020, 2022):** Recognized for impactful delivery on client-critical products.
- **Ace Award (2022):** For technical leadership in the team and mentoring new engineers.

VOLUNTEERING & LEADERSHIP

- Women Techmakers Ambassador (2023 - Present)
- President - Toastmasters Club (2021 - 2022)
- Mentor & Interviewer - Kongsberg Digital (2021 - 2025)

LANGUAGES

- English (Fluent)